Penny

A robot arm that plays pen-and-paper games with care home residents. She draws the board on real paper, reads every move with her camera, writes your moves when your hand isn't steady, learns the games you grew up with, and passes them on to other residents. Simulated SO-101 arm in MuJoCo; everything runs offline on one laptop.

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The need

~280k
people over 65 live in care homes in England and Wales[1]

1 in 4
older people are socially isolated[2], linked with 29% higher odds of early death[3]

~50%
of care home residents' waking time is spent inactive or dozing[4]

111k
unfilled posts in adult social care in England[5]

2.4m
older people barely use the internet, so a tablet isn't the answer[6]

07 Limits

Simulation only; grid and paper games; English only; she can't erase. Drawing is slow. The CNN has only seen simulator images. A carer loads paper.

08 Next steps

1 A paper roll for fresh sheets
2 Attachments for cards and board pieces
3 Onboard computer and mic: no laptop
4 The same code on a real SO-101

01 User and problem

Care home afternoons are long: staff are busy, other residents may not know the game, visits are short. Many residents loved paper games, but tremor, arthritis or a stroke now stops them writing (about 40% of stroke survivors keep arm problems; essential tremor affects 4–6% of over-65s[7]). Many family games were never written down, so no app has them.

Why it matters. In a meta-analysis of 411,000 older adults, those who used digital technology had lower odds of cognitive impairment[8]; Penny brings that kind of engagement to people who won't use a screen. Having a role matters too: volunteering and mentoring are linked with lower depression and longer life[9], which is why residents teach Penny their games.

02 Key features

Plays on real paper. Finds the A4 sheet, draws the board, reads every move with her camera and draws her own: noughts and crosses, hangman, dots and boxes, Footsteps and variants.

Writes for you. Say the square; she repeats it back and draws it, and refuses a taken or illegal square.

Learns your game. Explain it in conversation. She asks about anything missing, reads the rules back, solves the game, explains the winning idea, then draws and plays it.

Remembers and shares. A game library with records; taught games are offered to others.

Knows the session. Answers questions from live game state; difficulty from gentle to perfect play.

Adapts. Re-plans when the sheet moves mid-stroke; freezes when a hand comes over the paper.

03 System architecture

One Python process. Every move is read back from the paper before it counts. The language model understands people; the solver chooses moves; the camera checks everything. The language model never picks a move, a position or an arm motion.

04 Where each model runs

All on one laptop, no network. **Demo laptop: MacBook Air M5, 24 GB**; M5 rows measured on it with the demo running, M4 rows on a MacBook Air M4, 16 GB.

Component	Model or method	Runs on	Latency
Speech in	Silero VAD + whisper.cpp small.en	CPU/GPU	0.7–0.9 s a sentence (M4)
Understanding, questions	Gemma 4 E4B, Ollama	GPU	median 0.5–1.1 s, max 1.5 s, demo running (M5)
Teaching a game	Gemma 4 E4B + general solver	GPU, CPU	0.6–1.0 s a turn; solver under 2 s (M5)
Sheet finding	OpenCV with the camera model	CPU	2–25 ms a frame (M4)
Reading the board	CNN, 46.8k parameters, 48×48 crops	CPU	a few ms for 9 squares (M4)
Safety reflex	Depth image minus robot segmentation	CPU	within one camera frame, 0.1 s (M4)
Speech out	Piper, British female voice	CPU	46 ms to synthesise an 8-word sentence (M5)
Arm and drawing	MuJoCo 3, SO-101 model, pen-tip IK	CPU	grid ~40 s; one mark 12–15 s

05 Critical failure mode

Counting a move the resident didn't make. Ink can't be erased, so a wrong mark is permanent and confusing.

Unreadable mark (a squiggle, an "8"): she asks instead of guessing. Next: a yes counts only a new mark in an empty square on their turn; a no gets a fresh sheet, redrawn from the last legal state.

Misheard square: she repeats it before drawing; next, she waits for a yes first.

Misunderstood rules: she asks about gaps and reads the rules back before play. Below 90% confidence on any square, she asks.

06 Safety

Reflex layer. Depth image minus the robot's own body; anything over the paper freezes the arm until the area is clear for 1 s. Independent of the language model.

Gentle drawing. Spring-loaded felt tip; she lowers the pen until the touch sensor fires and holds 0.8 N; 30 mm/s while drawing.

Never lost. If the sheet moves mid-stroke: pen up, wait, find the sheet, redraw.

Private. Offline on the device; nothing leaves the room; no face recognition.

Sources 1 ONS (2023), care homes in 2021. 2 WHO, social isolation and loneliness. 3 Holt-Lunstad et al. (2015), 70 studies. 4 University of Kent, Ageing & Society. 5 Skills for Care (2024/25). 6 Age UK (2025). 7 Stroke and essential tremor reviews. 8 Benge and Scullin, Nature Human Behaviour (2025). 9 Jenkinson et al., BMC Public Health (2013). 8 and 9 are observational. Full list: docs/evidence.md.